Problem 5 - Skyline Delivery Fleet

Week 7 | Category B Assignment | Java solutions

Problem

DeliveryDrone and ScoutDrone share abstract Drone; only DeliveryDrone is Trackable. GroundRobot is Trackable but is not a Drone.

Complete Java Code

```
abstract class Drone {
    public abstract String fly();
}

interface Trackable {
    String getLocation();
}

class DeliveryDrone extends Drone implements Trackable {
    private final String id;
    public DeliveryDrone(String id) { this.id = id; }
    @Override public String fly() { return id + " flying with a parcel"; }
    @Override public String getLocation() { return id + " at Sector 4"; }
}

class ScoutDrone extends Drone {
    private final String id;
    public ScoutDrone(String id) { this.id = id; }
    @Override public String fly() { return id + " scouting the route"; }
}

class GroundRobot implements Trackable {
    private final String id;
    public GroundRobot(String id) { this.id = id; }
    @Override public String getLocation() { return id + " at Sector 4"; }
}

public class Main {
    static String getLocationIfTrackable(Object o) {
        if (o instanceof Trackable) {
            return ((Trackable) o).getLocation();
        }
        return "Tracking not available";
    }

    public static void main(String[] args) {
        System.out.println(getLocationIfTrackable(new DeliveryDrone("DR-1")));
        System.out.println(getLocationIfTrackable(new ScoutDrone("SC-1")));
        System.out.println(getLocationIfTrackable(new GroundRobot("GR-1")));
    }
}
```

Sample Output

```
DR-1 at Sector 4
Tracking not available
GR-1 at Sector 4
```

How it works

instanceof checks the capability interface, not the concrete class family. ScoutDrone has no Trackable implementation, while GroundRobot does even though it is unrelated to Drone.

Source basis: Week 7 Category B assignment/practice PDF. Examples and requirements are preserved; implementation details are filled in to create a complete runnable Java example.